

Panel: Barriers to Implementation of Accessibility Research in Industry Practice – and Vice Versa

Srishti Shekhar Agrawal
Computer Graphics Technology
Purdue University
West Lafayette, Indiana, USA
agraw219@purdue.edu

Amelia Laumann
SpoonStack
New York, New York, USA
amelia@spoonstack.org

Cynthia L Bennett
Google
New York, New York, USA
clbennett@google.com

Rua Mae Williams*
Computer Graphics and Technology
Purdue University
West Lafayette, Indiana, USA
rmwilliams@purdue.edu

Abstract

As public policy advances the rights of people with disabilities, and as corporations begin to recognize disabled people as core market segments, industry practice in accessible technology design and development improves day by day. Yet, as with other research domains there are lags and gaps in the implementation of accessibility between the lab and industry practice. In this panel, we describe multiple barriers to collaboration between academia and industry and how these barriers manifest as issues in implementation of research findings in industry and lack of adoption of best practices in academia, and vice versa. We then discuss how these specific cases complicate the imagined divide between academic and industry approaches to accessibility. How can notions of accessibility be expanded in both contexts to include overlooked dimensions like ethics, dark patterns, and cognition? This discussion moves towards more inclusive, impactful, and actionable accessibility practices across industry, academia, and public policy.

CCS Concepts

- **Social and professional topics** → Professional topics; Computing profession; User characteristics; People with disabilities;
- **Human-centered computing** → Accessibility; Accessibility theory, concepts and paradigms.

Keywords

Accessibility, Industry Practices, A11y, Translational Research

ACM Reference Format:

Srishti Shekhar Agrawal, Cynthia L Bennett, Amelia Laumann, and Rua Mae Williams. 2025. Panel: Barriers to Implementation of Accessibility Research in Industry Practice – and Vice Versa. In *Extended Abstracts of*

*Corresponding author.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/2025/04

<https://doi.org/10.1145/3706599.3716300>

the CHI Conference on Human Factors in Computing Systems (CHI EA '25), April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 5 pages.
<https://doi.org/10.1145/3706599.3716300>

1 Introduction

As recognition of disability and its impacts on human-computer interaction and technological experiences increases, so too does the research community's engagement and commitment to accessibility research. Likewise, as public policy advances the rights of people with disabilities, and as corporations begin to recognize disabled people as core market segments, industry practice in accessible technology design and development improves day by day. Yet, as with other research domains there are lags and gaps in the translation and implementation of accessibility between the lab and industry practice [2]. At the same time, working on these problems in daily contexts, industry practitioners have developed intimate knowledge of the practical realities of implementation that can be difficult to communicate back to researchers. This communication gap between academia and industry is rooted in disciplinary traditions and structural barriers to collaboration.

Academics are often motivated by career criteria for theoretical contributions, methodological rigor, and “high-risk, high reward” innovations well beyond the present practical limits in which industry operates. Industry prioritizes practical application, efficiency, and scalability, among other pragmatics more concerned with direct impact on existing consumers and user groups. Existing guidelines, such as W3C's WCAG, can be overwhelming and inaccessible to smaller teams or those without expertise, leading to gaps in adoption [2]. Personal background, exposure to accessibility issues, and organizational constraints further shape how accessibility is understood and implemented in both industry and academic contexts.

Prior research has focused on the educational context as a key site for intervention on the academic-industry divide [2, 10]. Such projects point out the vestigial nature of accessibility education in computing programs [2]. Infrequent, isolated, or “add-on” approaches to accessibility curriculum leave graduates without a deep understanding of accessibility issues and a shallow comprehension of accessibility as a core development practice. Scholars interested in this problem recognize that continuing education opportunities

are crucial to build industry capacity, and that focusing on traditional undergraduate and graduate pipelines is not enough [1]. While pedagogical reforms that center accessibility in computing education are essential to transforming industry practices [1, 2], this strategy is still limited in that it is one-directional (academic to industry). Furthermore, pedagogical and curricular reforms can only address industry lags in accessibility implementations that stem from a lack of skilled workers.

Failures of translation between academia and industry can be hard to identify, marked as they are by an absence of knowledge, practice, and material transfer rather than by any presence of a failed project or artifact. Still, we can perceive this rift when noting the volume of researcher-developed methods for more effective participation in the design process, such as ability-based design [15] and assets-based design [9], that don't appear to matriculate to industry standard practice in user evaluation. While industry practitioners may encounter these methods in targeted product development for specific categories of disability, their application to general product design and services remains absent. The Web Content Accessibility Guidelines may represent one of the few successful knowledge exchange projects, developed through the collaborative expertise of disabled technologists, researchers, and industry professionals [4]. Ironically, however, WCAG practice is under implemented both in industry and in academia, and many research artifacts fail to implement accessibility, thus excluding participation from disabled members of the general public [13].

Participatory design practice and strong, research-backed and user-evaluated guidelines are essential to developing an equitable global information ecosystem. However, recent projects have illustrated that gaps in guideline implementation, and indeed, gaps within the guidelines themselves, leave more than “access” on the line. Inaccessible information architecture can lead to devastating results, not only from the inability to access critical information, but also from the possibility that the information is itself physically dangerous [11, 12]. Recent work has identified how lapses in guideline implementation combine with the absence of meaningful enforcement and regulation to perpetuate physical harm through flashing content which can trigger nausea, migraines, seizures, and even death [14].

In this panel, we describe multiple barriers to collaboration between academia and industry and how these barriers manifest as issues in implementation of research findings in industry and lack of adoption of best practices in academia, and vice versa. We then discuss how these specific cases complicate the imagined divide between academic and industry approaches to accessibility. Are these gaps truly particular to academic and industry contexts, or is there something larger at play preventing institutions and industries alike from meaningful engagement with accessibility issues? What strategies can foster collaboration between academia's theoretical advancements, industry's practical constraints, and disabled people's lived realities? Finally, how can notions of accessibility be expanded in both contexts to include overlooked dimensions like ethical design, dark patterns, and cognitive load? By engaging the audience in this discussion, the panel aims to chart a path towards more inclusive, impactful, and actionable accessibility practices across industry, academia, and public policy.

2 Panelists

Each panelist brings a unique perspective on accessibility implementation and innovation issues across academia and industry. The organizer will introduce each panelist and then each panelist will describe their experiences with accessibility barriers from research, development, implementation, maintenance, and everything in between.

2.1 Rua M. Williams, Purdue University, Organizer

2.1.1 Bio. Dr. Williams is an Assistant Professor in User Experience Design at Purdue University and a former SSRC Just Tech Fellow (2022-2024). As PI of the CoLiberation Lab, Dr. Williams's work explores how disabled people imagine and build their own sociotechnical worlds, often in spite of existing structures of bias, stigma, and exclusion. Drawing from their own experiences as a disabled, neurodivergent, mad, and queer person, they also investigate how issues in technology policy and research practice interact to disrupt disabled people's bodily autonomy and access to meaningful public life.

2.1.2 Position Statement. When it comes to accessibility practices in technology design and deployment, my research typically focuses on methods for encouraging research reforms. For example, my recent Disability Theory in HCI Series has provided researchers a space to explore and engage with disability theory – critical disability studies literature, public scholarship, and activist's cultural artifacts. This series supports researchers to better understand how their choices in defining research inquiry and developing research protocols directly affect research outcomes and influence meaningful improvements in disabled people's daily lives. Much of my early publications put existing research in context of disability studies, demonstrating a severe lack of alignment between HCI research agendas and disabled people's own desires.

Through my work with Liz Jackson, a public scholar of disability design, I have attempted to bring the conversation of “misalignment” to industry audiences. Liz's concepts of the “Disability Don-gle” [5] and the “Wheelchair to Warfare Pipeline” [7], as well as critiques of inclusive design as a kind of extractive regime of erasure [6], are meant to support industry to reframe its relationship to disability and disabled users. These topics go beyond procedural notions of accessibility as a standards implementation problem and instead grapple with broader social beliefs that contribute to the sidelining and “ad hoc” nature of accessibility in research and practice.

For this panel, I am interested in unpicking the notion that Academia and Industry are truly separate spheres. While ostensibly, these different contexts have different priorities, agendas, and goals for knowledge-making, Academia and Industry are not distinct from broader society, and they are both subject to broader cultural narratives that influence attitudes toward and treatment of disabled people and disabled issues.

2.2 Srishti Agrawal, Purdue University, Panelist

2.2.1 Bio. Srishti Shekhar Agarwal is a UX Researcher with over four years of experience across multiple industries. She specializes

in data-driven research, accessibility advocacy, and the integration of AI to enhance accessibility. With a strong foundation in UX, engineering, and product management, Srishti creates holistic solutions that balance user needs, business goals, and technical feasibility. Srishti co-developed The Access Deck, a toolkit that helps UX designers integrate accessibility and inclusivity practices seamlessly into their design processes, making them second nature. Her research on responsible AI, particularly human values in RLHF datasets, was recognized with a spotlight paper at NeurIPS, and her design work received the Best Design Award at ACM Interaction Design and Children 2023. She has also delivered numerous talks on HCI, UX, and web accessibility. Passionate about bridging the gap between academia and industry, Srishti's current research explores how AI can enhance project-based learning, offering insights that benefit both academic curricula and industry practices. She advocates for embedding accessibility throughout the design and development lifecycle to foster greater inclusion.

2.2.2 Position Statement. Accessibility should be a core, inherent aspect of design, not something justified based on its appeal to non-disabled users. Truly accessible design enhances the experience for all users, aligning with universal design principles that place inclusion at the heart of the process. When interviewing practicing UX professionals, I found that many of them believe that they do disability accessibility in their daily work, and yet they cannot articulate what accessibility design practice might be. However, I've also seen how accessible frameworks and tools can demystify complex guidelines, driving adoption even within resource-constrained teams. For example, I developed a project called The Access Deck, in which the complex and multi-layered accessibility guidelines are turned into a kind of introspective card game which designers can play alone or in teams to think through their designs and witness, with concrete examples and implementation recommendations, how their designs might fail to uphold these guidelines. These efforts demonstrate the power of clear, practical solutions that integrate accessibility seamlessly into the design lifecycle.

I advocate for expanding the scope of accessibility, going beyond traditional definitions to include ethical design, cognitive load, and dark patterns—elements that profoundly impact user experiences. In this context, accessibility is not just about meeting compliance but about creating environments where all users, regardless of ability, can engage meaningfully. As technology and AI continue to evolve, inclusivity will only become more essential. The design community must embrace the responsibility of embedding accessibility into every stage of development—from ideation to implementation—ensuring it is prioritized and not just considered. My work aims to advance these principles across academia, industry, and public domains, fostering a culture where accessibility and inclusivity are shared responsibilities and default design practices.

2.3 Amelia Laumann, SpoonStack, Panelist

2.3.1 Bio. Amelia Laumann (she/her/hers) is a software industry veteran of over two decades, having worked in a wide range of roles and across many different facets of digital technology. In addition to her substantial work in computer programming, she has held leadership positions in notable software shops, including several years as Director of Technology. Throughout her career, Amelia

has focused on cross-discipline and cross-cultural techniques as a powerful way to improve organizational effectiveness, technical rigor, and collaborative efficiency. Her background in both applied development and business strategy, combined with her lived experiences of invisible disability, gives her a unique perspective on the challenges that impede efforts for technological accessibility. She is the founder of SpoonStack and creator of a novel approach to organizing and conducting software creation, called Disability-Driven Development. This initiative synthesizes knowledge and practical wisdom from many different fields, including social justice work, to inform possible futures of equity and access for digital technology.

2.3.2 Position Statement. There is absolutely no shortage of obstacles when it comes to pushing for more accessible technologies. These can range from practical, almost mundane issues - like funding and scheduling - to more subtle and difficult-to-recognize frictions that are rooted in the societal and cultural patterns and systems we exist within. Like many disability justice practitioners before me, I contend that underneath the often technical or pragmatic surface layer, these challenges are in fact human problems.

The uncomfortable reality is that our institutions and corporations do not consistently prioritize accessibility work because on a cultural level we do not prioritize the quality-of-life of disabled people. Both academia and industry have long-held traditional incentive structures and success metrics that directly conflict with such prioritization. Both emerged historically from a larger sociopolitical reality in which the experiences of our disabled population are rarely placed at the forefront of our collective consideration. These priorities are consistently reflected even in non-profit and volunteer software and tech development efforts, such as the open-source software ecosystem. Unless we make an intentional effort to reckon with the non-technological roots of systemic and cultural ableism, we will have only limited success in meaningfully addressing access inequity within the realm of technology itself.

I believe that the most reliable way to overcome the practical hurdles of accessibility work is to learn from other historical social justice movements. Movements for gender equity, racial equity, LGBTQ+ equity, and so on are rich with lessons about what does (and does not) work when addressing such large-scale challenges. Human relationships can be key for shifting resource allocations and for motivating the successful creation of infrastructure and policy that upholds inclusivity and equity goals. By centering the experiences and perspectives of disabled people, and ensuring they are placed as leaders and strategists rather than merely “participants” or expert consultants, we can grow networks of connections that substantially alter the status quo.

2.4 Cynthia L. Bennet, Affiliation, Panelist

2.4.1 Bio. Cynthia Bennett is a senior research scientist at Google Research. Her research focuses on understanding the impacts of generative AI on people with disabilities, and informing AI development to respectfully represent and be accessible to them. More broadly her research concerns sociotechnical systems, ethics, and disabled, worldbuilding. Her dissertation research uncovered ableist rhetorics in inclusive and empathic design tools which figure users in need of professionalized technologists' expertise. Meanwhile, people with disabilities have design expertise and contribute

to technological knowledge which often goes un[der]recognized. She is a disabled woman scholar committed to raising participation of people with disabilities in HCI research and the tech industry.

2.4.2 Position Statement. As companies integrate accessibility compliance into product development cycles, accessibility becomes more of the everyday parlance. I find the elevation of accessibility into core development cycles exciting. As a researcher, I am identifying domains for which we do not have established accessibility standards, and I question the boundaries of accessibility itself. People with disabilities are not being represented by AI in the ways that disabled people would like to be represented [8]. At the same time, people with disabilities are increasingly using general purpose AI to seek information. There are major quality differences in generative AI output related to accessibility and disability queries [3]. Accessibility is much more than UI design. The quality and reliability of accessibility and disability information, and disability representation are crucial issues for our changing information infrastructure. I argue that higher level guidelines which have become the foundation of how accessibility is incorporated in product development must be continually updated and supplemented. Translating these research findings into guidance and witnessing its implementation is an open question.

As a disabled person who researches accessibility and disability, I pay attention to who is in the room and why people think I am there. I am grateful that I work in an environment where people generally recognize that including the target population in development is ideal. Inclusion is often conflated with mere invitation to participate in user studies.. While disabled representation in user studies is essential to effective product design, there are other mechanisms for integrating disabled expertise into practice. Disabled employees are producers of knowledge, expertise, and labor. Sometimes, I feel tensions in the motivations for my inclusion – I often suspect that my being disabled may be the reason I was invited, rather than the merits of my research. Accounting for such dynamics when we develop best practices—whether for translating across institutional boundaries or implementing research findings into product—will help disabled people to get the recognition we deserve and allow us to pursue our own interests beyond the “disabled disability expert”.

3 Discussion Format

The organizer will lead the panel in discussion on key topics, with opportunities for audience interaction. The panel will take a total of 90 minutes: 5 minutes for introductions led by the panel organizer, followed by four 15-minute rounds (60 minutes) of discussion based on organizer questions to address the following topic areas. (Each of the three participants will have approximately 3 minutes each for responses per question, leaving 6 minute buffers.) The final 25 minutes are reserved for open Q & A.

3.1 Differences Between Academia’s and Industry’s approach to Accessibility:

In academia, accessibility is often constructed as an evolving field of study focused on developing innovative theories, comprehensive frameworks, or cutting-edge technology innovations. Researchers investigate the broader impacts of accessibility on user experience, with in-depth studies that push the boundaries of what is considered

accessible. However, industry teams may prioritize functionality, efficiency, and timeliness. Often, individual companies may not have a specific focus on disabled users, treating accessibility as a compliance checkbox or an afterthought rather than a core design feature.

Discussion prompt: How do academic approaches to accessibility differ from industry practices, and what accounts for these differences? What structural or cultural differences between academia and industry contribute to these divergent views on accessibility? How can academia adapt its insights to make them more relevant to industry’s time and resource constraints? How can industry reframe its approach to accessibility to move beyond compliance and towards innovation in inclusive design?

3.2 The Influence of Personal Background on Accessibility Practices:

Individual exposure to accessibility—through personal experiences or interactions—often informs a designer’s awareness and commitment to accessibility. Personal connections with disability can drive designers to view accessibility not just as a requirement, but as a valuable practice for all users.

Discussion prompt: How do personal backgrounds and lived experiences shape one’s approach to accessibility, and how can this perspective be shared to build stronger commitments to accessibility within design teams? How can exposure to accessibility issues be shared or stimulated within teams to build capacity in accessibility practice? What role does empathy play in shaping effective accessibility practices? If empathy is not the right frame for behavior change, what is?

3.3 Barriers to Robust Implementation of Existing Guidelines:

Current accessibility guidelines are thorough, covering various compliance aspects, but their extensive, text-heavy nature makes them difficult for many teams—especially smaller ones—to navigate. Teams without a dedicated accessibility expert may find these guidelines intimidating, overwhelming, and challenging to implement practically.

Discussion prompt: How can existing accessibility guidelines and policies, such as WCAG, ADA, and Section 508, be made more actionable for design teams? What adaptations to WCAG guidelines, ADA compliance guides, or other guidelines could make them more designer-friendly and easier to implement? How can tangible or digital tools help designers better integrate these guidelines into everyday workflows? What specific challenges do small and mid-sized companies face in implementing accessibility measures compared to large tech firms?

3.4 Holistic Approaches to Accessibility: Expanding Beyond Traditional Topics

Accessibility is often narrowly defined, yet related issues—such as ethical design, dark patterns, cognitive load, and mental well-being—are critical components of inclusive design that affect a broad range of users. Incorporating these aspects can lead to a

more holistic approach that not only meets accessibility needs but also promotes ethical, user-centered design.

Discussion prompt: What broader topics should be included under the umbrella of accessibility and inclusivity? Should accessibility be expanded to encompass areas like ethical design, dark patterns, and user cognitive load? How can these broader accessibility considerations be practically incorporated into design processes?

4 Outcomes

Attendees of this panel will witness and engage in conversation that explores the space “between” Academia and Industry and how broader social issues influence research outcomes and best practices in both spheres. Panellists will support attendees in blurring the boundaries between Academia and Industry and recognizing these as concepts and constructs that serve a rhetorical purpose that is not always aligned with reality. Crucially, while there are fundamental differences in the experience of working in industry vs. in a lab, as accessibility researchers and practitioners, we have more to gain from exploring the shared sources of barriers to effective access so that we can address these barriers together.

References

- [1] Andy Coverdale, Sarah Lewthwaite, Yasmine N. Elglaly, Scott Hollier, Sarah Horton, Kate Sonka, and Gottfried Zimmermann. 2024. Editorial: Advancing digital accessibility in academic and workplace education. *Front. Comput. Sci.* 6, (May 2024). <https://doi.org/10.3389/fcomp.2024.1419986>
- [2] Andy Coverdale, Sarah Lewthwaite, and Sarah Horton. 2022. Teaching accessibility as a shared endeavour: building capacity across academic and workplace contexts. In *Proceedings of the 19th International Web for All Conference (W4A '22)*, April 27, 2022. Association for Computing Machinery, New York, NY, USA, 1–5. <https://doi.org/10.1145/3493612.3520451>
- [3] Maitraye Das, Alexander J. Fiannaca, Meredith Ringel Morris, Shaun K. Kane, and Cynthia L. Bennett. 2024. From Provenance to Aberrations: Image Creator and Screen Reader User Perspectives on Alt Text for AI-Generated Images. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*, May 11, 2024. Association for Computing Machinery, New York, NY, USA, 1–21. <https://doi.org/10.1145/3613904.3642325>
- [4] W3C Web Accessibility Initiative (WAI). Introduction to Web Accessibility. *Web Accessibility Initiative (WAI)*. Retrieved April 19, 2024 from <https://www.w3.org/WAI/fundamentals/accessibility-intro/>
- [5] Liz Jackson, Alex Haagaard, and Rua M. Williams. 2022. Disability Dongle. *CAS-TAC Platypus Blog*. Retrieved January 26, 2023 from <https://blog.castac.org/2022/04/disability-dongle/>
- [6] Liz Jackson and Rua M. Williams. 2023. The Fashion Empire Built on Stolen Ideas. *The New Republic*. Retrieved December 4, 2024 from <https://newrepublic.com/article/176381/runway-dreams-fashion-empire-stolen-ideas>
- [7] Liz Jackson and Rua M. Williams. 2024. The Wheelchair to Warfare Pipeline: How Disabled People Get Exploited to Build the Technology of War. *The New Republic*. Retrieved from <https://newrepublic.com/article/179391/wheelchair-warfare-pipeline-disability-technology>
- [8] Kelly Avery Mack, Rida Qadri, Remi Denton, Shaun K. Kane, and Cynthia L. Bennett. 2024. “They only care to show us the wheelchair”: disability representation in text-to-image AI models. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*, May 11, 2024. Association for Computing Machinery, New York, NY, USA, 1–23. <https://doi.org/10.1145/3613904.3642166>
- [9] Lucy Pei and Bonnie Nardi. 2019. We Did It Right, But It Was Still Wrong: Toward Assets-Based Design. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems (CHI EA '19)*, May 02, 2019. Association for Computing Machinery, New York, NY, USA, 1–11. <https://doi.org/10.1145/3290607.3310434>
- [10] Kate Sonka, Casey McArdle, and Liza Potts. 2021. Finding a Teaching A11y: Designing an Accessibility-Centered Pedagogy. *IEEE Trans. Prof. Commun.* 64, 3 (September 2021), 264–274. <https://doi.org/10.1109/TPC.2021.3091190>
- [11] Laura South. 2022. Improving Online Accessibility for Individuals with Photosensitive Epilepsy. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems (CHI EA '22)*, April 28, 2022. Association for Computing Machinery, New York, NY, USA, 1–4. <https://doi.org/10.1145/3491101.3503805>
- [12] Laura South, Caglar Yildirim, Amy Pavel, and Michelle A. Borkin. 2023. Exploratory Thematic Analysis of Crowdsourced Photosensitivity Warnings. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 19, 2023. Association for Computing Machinery, New York, NY, USA, 1–7. <https://doi.org/10.1145/3544549.3585649>
- [13] Rua M. Williams and Juan E. Gilbert. 2019. “Nothing about us without us”: Transforming participatory research and ethics in human systems engineering. In *Advancing Diversity Inclusion and Social Justice Through Human Systems Engineering*, Rod Roscoe and Erin Chiou (eds.). Taylor & Francis Group.
- [14] Rua Mae Williams, Chorong Park, Laila Sameer Dodhy, Monaami Pal, Atharva Anand Dnyanmote, Luchcha Angel Lam, and Sean Joo. 2024. “Not Only Annoying but Dangerous”: devising an ecology of protections for photosensitive social media users. In *Proceedings of the 26th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '24)*, October 27, 2024. Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3663548.3675610>
- [15] Jacob O. Wobbrock, Krzysztof Z. Gajos, Shaun K. Kane, and Gregg C. Vanderheiden. 2018. Ability-based design. *Commun. ACM* 61, 6 (May 2018), 62–71. <https://doi.org/10.1145/3148051>